

A Comprehensive Survey of Pretraining Tasks in Modern Artificial Intelligence

1. Introduction: The Pretraining Paradigm

The development of modern Artificial Intelligence (AI) has been profoundly shaped by a paradigm known as pretraining. This initial, foundational stage of model development has become the cornerstone of today's most capable AI systems, particularly the class of models known as foundation models. Pretraining has evolved from a simple initialization strategy into a resource-intensive, industrial-scale process that imbues models with a general understanding of data, which is then refined for specialized applications. This introductory chapter defines the pretraining paradigm, contextualizes its role in the era of foundation models, charts its evolution from supervised to self-supervised learning, and outlines the structure of this comprehensive survey.

1.1 Defining Pretraining

Pretraining is the process of training a machine learning model, typically a deep neural network, on a large-scale, general-purpose dataset before it is adapted to a specific downstream task.¹ This initial phase is designed to equip the model with a broad and foundational understanding of the patterns, structures, syntax, and semantics inherent in a particular data modality.³ For instance, a language model might be pretrained on a vast corpus of text encompassing trillions of words from books, articles, and websites to learn grammar, facts about the world, and reasoning abilities.³ Similarly, a computer vision model can be pretrained on millions of images to learn to recognize fundamental visual features like edges, textures, shapes, and object parts.⁴

The core objective of pretraining is to learn a set of model parameters—the weights and biases of the neural network—that serve as a powerful and informed starting point for subsequent learning.⁵ Instead of beginning the training process for a specific application with randomly initialized parameters, a pretrained model starts with a rich, learned representation of the data.⁷ This process is analogous to providing a general education; the model becomes a versatile generalist, ready to be specialized for a multitude of specific roles through a subsequent process known as fine-tuning.³ This foundational knowledge makes the model more efficient and powerful, significantly reducing the amount of task-specific data and computational resources required to achieve high performance on downstream applications.³

1.2 The Role of Pretraining in the Age of Foundation Models

The advent of foundation models—large-scale models such as BERT, GPT-4, and CLIP that are pretrained on massive, diverse datasets—has solidified pretraining as the central process in state-of-the-art AI development.⁹ These models serve as a reusable base upon which a wide array of specialized AI applications can be built.¹⁰ The remarkable capabilities of foundation models, including their ability to generalize to new tasks and perform transfer learning with high efficacy, are a direct result of the knowledge acquired during the pretraining phase.⁵

Pretraining is distinct from, yet intrinsically linked to, the concept of fine-tuning. The two represent sequential stages in a dominant AI development workflow.¹²

- **Pretraining:** This is the initial, computationally intensive stage where the model learns general features from a vast, often unlabeled, dataset. It is during this phase that the model's foundational knowledge is established.⁴
- **Fine-tuning:** This is the subsequent, comparatively lightweight stage where the pretrained model is further trained on a smaller, task-specific, and typically labeled dataset. This process adapts the model's general knowledge to the nuances of the specific application, such as sentiment analysis in text or medical image segmentation in vision.⁵

The pretraining phase provides a crucial "head start," an inductive bias that makes the fine-tuning process significantly more effective and efficient.² By leveraging the rich representations learned during pretraining, models can achieve high performance on downstream tasks with far less specific data than would be required if training from a

random initialization.⁴ This two-stage paradigm has become the standard for developing high-performance AI systems, enabling organizations to leverage the power of foundation models without incurring the prohibitive cost of pretraining them from scratch.¹³

1.3 The Paradigm Shift: From Supervised to Self-Supervised Learning

The methodology of pretraining has undergone a significant evolution. Early successes, particularly in computer vision, were dominated by **supervised pretraining**. The standard practice involved training a model on a large, manually annotated dataset, with ImageNet being the canonical example.⁴ In this task, a model learns to classify images into one of a thousand predefined categories. The features learned in this process, especially in the network's earlier layers, proved to be highly transferable to a variety of other vision tasks.¹⁴

However, the supervised approach is fundamentally limited by the availability of high-quality labeled data. The process of manual annotation is expensive, time-consuming, and difficult to scale, creating a bottleneck for model development.¹⁷ This limitation spurred a pivotal shift towards

self-supervised learning (SSL), a paradigm that has come to define modern pretraining.⁴ SSL enables models to learn from vast quantities of unlabeled data by creating supervisory signals directly from the data itself.²⁰ This is achieved by devising "pretext tasks," where the model is trained to solve a problem for which the labels can be automatically generated. For example, a model might be tasked with predicting a word that has been artificially removed from a sentence or reconstructing a part of an image that has been masked out.¹⁸ This innovation unlocked the potential of the internet's massive, unstructured data repositories, which are orders of magnitude larger than any existing labeled dataset.²²

This methodological shift has been so profound that it has altered the very lexicon of the field. What was once simply called "training" a model on a large dataset is now almost universally referred to as "pretraining".²³ This linguistic evolution reflects a fundamental change in the AI development lifecycle. In the past, a model was typically trained end-to-end on a specific task's dataset.²⁴ With the rise of transfer learning, a two-stage process emerged, but the first stage was still just "training." The advent of models like BERT and GPT, however, transformed this initial stage into a massive,

self-supervised, and computationally demanding endeavor, often requiring thousands of GPUs and months of training time.²⁴ In contrast, the second stage, fine-tuning, became a relatively lightweight and accessible process.²⁶ The term "pretraining" was adopted to signify this new reality: the "pre-" now implies not just a temporal precedence but a distinct, foundational, and often separately executed phase of development.²³ This bifurcation of the AI pipeline into industrial-scale pretraining and application-specific fine-tuning has been a key driver in democratizing access to cutting-edge AI capabilities.¹⁵ A pretrained model can be understood as a highly compressed, implicit representation of the statistical patterns and knowledge contained within its massive training corpus. The self-supervised objectives compel the model to learn the correlations, structures, and regularities of the data, encoding this information into its parameters.² Consequently, the capabilities, limitations, and inherent biases of a foundation model are a direct reflection of the data it was pretrained on, making the curation and quality of the pretraining dataset a matter of paramount importance.³⁰

1.4 Scope and Structure of this Survey

This survey provides an exhaustive investigation into the pretraining tasks that form the foundation of modern AI across a comprehensive range of modalities. It moves beyond a simple catalog of methods to offer a nuanced understanding of their underlying mechanisms, their conceptual connections, and their evolutionary trajectory. The report is structured as follows:

- **Section 2** establishes a formal taxonomy of pretraining objectives, creating a conceptual framework for comparing the diverse tasks discussed.
- **Sections 3, 4, and 5** provide in-depth technical analyses of pretraining tasks specific to the core modalities of Natural Language Processing, Computer Vision, and Speech/Audio Processing, respectively.
- **Section 6** explores the domain of Multimodal Learning, focusing on tasks designed to align representations from different data types, particularly vision and language.
- **Section 7** examines the application of pretraining principles to specialized and emerging domains, including Graph Representation Learning and Reinforcement Learning for Robotics.
- **Section 8** concludes the survey by synthesizing the key evolutionary trends and exploring the future of pretraining, with a particular focus on the development of

world models for planning and decision-making.

Through this structured approach, this report aims to provide a definitive and insightful reference on the pretraining tasks that are driving progress across the landscape of artificial intelligence.

2. A Taxonomy of Pretraining Objectives

To systematically navigate the diverse landscape of pretraining tasks, it is essential to establish a clear taxonomy based on the nature of the learning objective and the type of supervisory signal employed. Pretraining methods can be broadly categorized into two main paradigms: supervised pretraining, the classical approach reliant on human-annotated labels, and self-supervised learning (SSL), the modern paradigm that generates supervision from the data itself. Within SSL, tasks can be further classified into three primary families—contrastive, generative, and predictive—each embodying a distinct philosophical approach to representation learning.

2.1 Supervised Pretraining

Supervised pretraining represents the traditional approach to transfer learning, where a model is first trained on a large-scale, manually labeled dataset before being fine-tuned for a different, often smaller, downstream task. For nearly a decade, this was the dominant pretraining paradigm in computer vision.¹⁶

The canonical example of this approach is **Image Classification on ImageNet**. The ImageNet Large Scale Visual Recognition Challenge (ILSVRC) dataset contains over 1.2 million images, each meticulously labeled with one of 1000 object categories.³² By training a model, typically a Convolutional Neural Network (CNN), to perform this classification task, the network is forced to learn a hierarchy of visual features. Early layers learn to detect simple patterns like edges and colors, intermediate layers learn to recognize more complex textures and parts of objects, and deeper layers learn to identify entire objects.¹⁴ This rich hierarchy of learned features proved to be highly transferable, providing a powerful initialization that significantly improved performance and accelerated convergence on a wide array of other vision tasks, from

object detection to semantic segmentation.¹⁴

Despite its historical success, the supervised pretraining paradigm faces significant limitations. The primary bottleneck is its reliance on human-annotated data, a process that is notoriously expensive, labor-intensive, and difficult to scale.¹⁷ Furthermore, the learned representations can become overly specialized to the specific classes and biases present in the pretraining dataset, which can limit their generalization to new and unseen tasks.³³ Research has also shown that as the size of the downstream dataset and the strength of data augmentation techniques increase, the benefits of supervised ImageNet pretraining can diminish or even become negligible.¹⁶ These challenges, coupled with growing concerns over the ethical and copyright status of large, web-scraped labeled datasets, have motivated a widespread shift towards self-supervised alternatives.³¹

2.2 Self-Supervised Learning (SSL): The Modern Paradigm

Self-Supervised Learning (SSL) has emerged as the dominant paradigm for pretraining modern foundation models. SSL is a form of unsupervised learning where the supervisory signal for a task is generated from the input data itself, rather than from external, human-provided labels.¹⁷ This is achieved by creating a "pretext task" (also known as a proxy or auxiliary task), which is a problem the model must solve that is not the ultimate downstream task but serves as a means to learn a rich and useful data representation.¹⁷

The fundamental advantage of SSL is its ability to leverage the virtually limitless supply of unlabeled data available in the world, such as text from the internet or videos from public platforms.¹⁸ By learning to solve a well-designed pretext task, the model is forced to capture the underlying structure, semantics, and statistical regularities of the data. The resulting representations are often more general and robust than those learned from supervised pretraining on a finite set of classes.³³ SSL objectives can be broadly organized into three major families, which form the basis of our taxonomy.

2.3 Family 1: Contrastive Methods

Contrastive methods are based on the principle of learning by comparison. The core objective is to train a model to map the data into an embedding space where representations of semantically similar samples are pulled closer together, while representations of dissimilar samples are pushed farther apart.¹⁸ The model learns a discriminative function that scores the similarity between pairs of inputs.³⁵

In the context of SSL, "similar" or "positive" pairs are typically generated through data augmentation. For example, two different random crops of the same image are considered a positive pair. "Dissimilar" or "negative" pairs are simply other samples in the training batch or a specially maintained memory bank.¹² The model is then trained with a contrastive loss function, such as InfoNCE, which encourages the similarity score of positive pairs to be high relative to the scores of all negative pairs.³⁷

This approach has been exceptionally successful across multiple modalities.

- **In Vision:** Models like **SimCLR** and **MoCo** learn powerful visual representations by contrasting different augmented views of images.³⁸
- **In Multimodal Learning:** Models like **CLIP** learn to align images and text by treating matching image-caption pairs as positives and all other combinations in a batch as negatives.⁴⁰
- **In Speech:** **wav2vec 2.0** uses a contrastive task to identify the correct quantized speech representation for a masked time step from a set of distractors.⁴¹

2.4 Family 2: Generative Methods

Generative methods operate on the principle of learning by reconstruction or synthesis. These approaches typically involve corrupting the input data in some manner and then training the model to restore the original, uncorrupted version.¹⁷ This forces the model to learn a deep and holistic understanding of the data's structure and content.

This paradigm often employs an encoder-decoder architecture. The encoder learns to compress the (uncorrupted or partially visible) input into a rich, latent representation. The decoder then uses this representation to generate or reconstruct the original signal.⁴² To succeed at this task, the latent representation must capture all the information necessary for the reconstruction, thus encouraging the formation of

high-quality features.

Generative pretraining tasks have proven to be highly effective and scalable.

- **In Vision: Masked Autoencoders (MAE)** corrupt an image by masking a large percentage of its patches and train the model to reconstruct the missing pixels.⁴² Earlier methods used pretext tasks like **image inpainting** (filling in a missing rectangular region) and **image colorization** (predicting color from a grayscale image).⁴³
- **In NLP:** The **T5** model's pretraining objective involves reconstructing corrupted "spans" of text that have been replaced by a sentinel token, a task known as a denoising objective.³⁰

2.5 Family 3: Predictive (or Auxiliary) Methods

Predictive methods, which often overlap with generative ones, are based on the principle of learning by predicting a withheld property or part of the data. The distinction is often that the target of the prediction is a discrete property or label rather than a high-dimensional continuous signal like pixels. A pretext task is designed where some attribute of the data is used to generate a pseudo-label, which the model is then trained to predict.¹⁷

This is the foundational paradigm for most large language models and has seen significant success in speech processing as well.

- **In NLP: Masked Language Modeling (MLM)**, the core of BERT, involves predicting the original identities of randomly masked tokens in a sentence.⁴⁴ **Causal Language Modeling (CLM)**, the basis for GPT, involves predicting the next token in a sequence.⁴⁵ In both cases, the targets are discrete tokens from a vocabulary.
- **In Speech: HuBERT** trains a model to predict discrete cluster IDs for masked segments of an audio signal. These cluster IDs are generated beforehand by running a k-means algorithm on the audio features, serving as pseudo-labels.⁴⁶
- **In Vision:** Early SSL work explored predictive tasks like predicting the degree of rotation applied to an image or solving a jigsaw puzzle by predicting the correct relative positions of shuffled image patches.¹⁸

The following table provides a consolidated overview of these major pretraining tasks,

organized according to this taxonomy. It serves as a high-level map of the field, which will be explored in detail in the subsequent chapters.

Pretraining Task	Primary Modality	Learning Paradigm (Taxonomy)	Core Objective	Seminal Model(s)
Image Classification	Vision	Supervised	Predict the correct object class from a fixed set of labels.	AlexNet, VGG, ResNet
Causal Language Modeling (CLM)	Language	Predictive	Predict the next token in a sequence given the preceding tokens.	GPT series
Masked Language Modeling (MLM)	Language	Predictive	Predict randomly masked tokens in a sequence using bidirectional context.	BERT
Replaced Token Detection (RTD)	Language	Predictive (Discriminative)	Distinguish original tokens from plausible fakes generated by another model.	ELECTRA
Text-to-Text Denoising	Language	Generative	Reconstruct corrupted spans of text in a unified input-output format.	T5, BART
Contrastive Learning	Vision, Speech	Contrastive	Maximize agreement between	SimCLR, MoCo

			augmented views of the same sample.	
Masked Image Modeling (MIM)	Vision	Generative	Reconstruct masked patches of an image (pixels or discrete tokens).	MAE, BEiT
Generative Pretext Tasks	Vision	Generative	Reconstruct a property of the image, e.g., colorization, inpainting.	Context Encoders
Masked Prediction of Audio Units	Speech, Audio	Predictive/Contrastive	Predict discrete representations for masked segments of an audio signal.	wav2vec 2.0, HuBERT
Vision-Language Contrastive (VLC)	Multimodal	Contrastive	Align image and text representations by contrasting matched/unmatched pairs.	CLIP, ALIGN

A notable trend revealed by this taxonomy is the convergence of pretraining objectives across different modalities. Initially, SSL methods were highly specific to their domain; vision tasks centered on geometric properties like rotation and patch positions, while NLP focused on sequential token prediction.¹⁸ The widespread adoption of the Transformer architecture, which can process diverse data types as sequences of tokens or patches, served as a unifying force.⁴⁷ This architectural convergence facilitated a conceptual convergence of pretraining tasks. The "mask and predict" strategy, which proved transformative for NLP in BERT, was successfully adapted to vision in MAE and to audio in HuBERT.⁴² Similarly, contrastive learning, which gained prominence in vision with SimCLR and MoCo, became a core component of speech models like wav2vec 2.0 and the dominant paradigm for aligning modalities in vision-language models like CLIP.³⁸ This suggests that the fundamental principles of self-supervision—learning by predicting missing information

and learning by contrasting different views—are proving to be remarkably universal. The primary challenge is often modality-specific: defining the appropriate way to "tokenize" the data and design the "masking" or "augmentation" strategy.

Furthermore, the dichotomy between contrastive and generative/reconstructive methods reflects a fundamental trade-off in the type of knowledge being learned. Contrastive methods excel at learning highly discriminative features, which is why models like SimCLR perform exceptionally well on linear probe evaluations, a benchmark for feature quality.³⁸ The objective explicitly forces the model to learn what makes different instances unique. In contrast, generative methods like MAE, which require the reconstruction of a high-dimensional signal (e.g., pixels) from a partial input, compel the model to develop a more holistic and detailed understanding of the data's internal structure.⁴² This difference is reflected in downstream performance; MAE pretraining has proven exceptionally effective for fine-tuning on dense prediction tasks like semantic segmentation, which benefit from its rich spatial understanding, even if its linear probe performance is not as high as top contrastive methods.⁴² Therefore, the choice of pretraining objective family is not merely a matter of efficiency but a strategic decision about the nature of the representations one wishes to instill in the model, with contrastive methods favoring discriminative power and generative methods favoring holistic, reconstructive knowledge. This has naturally led to the exploration of hybrid approaches that seek to combine the strengths of both paradigms.⁵⁰

3. Pretraining Tasks in Natural Language Processing

The field of Natural Language Processing (NLP) has been revolutionized by the advent of large-scale pretraining on Transformer-based architectures. The pretraining tasks developed for this domain are designed to instill a deep understanding of language—from syntax and grammar to semantics and world knowledge—by leveraging vast, unlabeled text corpora. This section provides a detailed technical examination of the seminal pretraining objectives that have defined the modern era of NLP and Large Language Models (LLMs).

3.1 Masked Language Modeling (MLM): Learning Bidirectional Context

Masked Language Modeling (MLM) is a self-supervised pretraining objective introduced with the Bidirectional Encoder Representations from Transformers (BERT) model.⁴⁴ Its key innovation was to enable the learning of deep bidirectional representations, meaning the model learns to understand a word based on its full context, including the words that come both before and after it. This was a significant departure from previous state-of-the-art models like GPT, which were unidirectional and could only consider the left-side (preceding) context.⁵²

The mechanism of MLM is as follows:

1. **Input Corruption:** A certain percentage of the input tokens in a sentence, typically 15%, are randomly selected for prediction.⁴⁴
2. **Masking Strategy:** To prevent a mismatch between the pretraining phase (where the model sees a special token) and the fine-tuning phase (where it does not), a specific masking strategy is employed. Of the selected tokens, 80% are replaced with the token, 10% are replaced with a random token from the vocabulary, and the remaining 10% are left unchanged.⁴⁴
3. **Prediction Objective:** The model, which uses a multi-layer Transformer encoder architecture, is then tasked with predicting the original identity of these selected tokens based on the surrounding unmasked context.⁵⁴ Because the Transformer encoder's self-attention mechanism allows every token to attend to every other token in the sequence, the prediction for a masked position is informed by the entire sentence, thus making the representation bidirectional.⁵²

In its original formulation, BERT was also pretrained with an auxiliary task called **Next Sentence Prediction (NSP)**. This was a binary classification task where the model was given two sentences, A and B, and had to predict whether B was the actual sentence that followed A in the original corpus or a random sentence.⁴⁴ The goal was to improve performance on downstream tasks that require understanding relationships between sentences, such as question answering. However, subsequent research and models like RoBERTa demonstrated that the NSP task was of limited utility and could even be detrimental to performance, leading many later models to abandon it in favor of training on longer contiguous text sequences.⁴⁴

The impact of MLM was transformative. By learning deep, context-aware representations, BERT and its derivatives established new state-of-the-art performance across a wide range of natural language understanding (NLU) benchmarks, including text classification, named entity recognition, and question

answering.⁴⁴

3.2 Causal Language Modeling (CLM): Learning to Generate

Causal Language Modeling (CLM), also known as autoregressive language modeling or next-token prediction, is the fundamental pretraining task for generative models like the Generative Pre-trained Transformer (GPT) series.⁴⁵ The objective is conceptually simple: given a sequence of tokens, the model is trained to predict the very next token in the sequence.⁵⁶

The mechanism relies on a "causal" or "masked" self-attention pattern within the Transformer decoder architecture. When predicting the token at a given position, the model is only allowed to attend to the tokens that came before it (and the current token itself); it is prevented from "seeing" any future tokens in the sequence.⁵³ This unidirectional constraint is what makes the model autoregressive and inherently suited for generation. During training, the model is presented with vast amounts of text and learns to predict the next word at every single position in the corpus, effectively learning the statistical patterns of language.⁴⁵

The primary strength of CLM is its natural aptitude for text generation. Once pretrained, a CLM-based model can generate coherent and contextually relevant text by iteratively predicting the next token, appending it to the sequence, and feeding the new sequence back into the model as input.⁵³ This capability is the foundation for applications such as chatbots, content creation, summarization, and dialogue systems.⁴⁵ The remarkable emergent abilities of modern LLMs, such as in-context learning—the ability to perform tasks with zero, one, or a few examples provided in the prompt without any weight updates—are a direct consequence of scaling up models pretrained with the CLM objective to immense sizes.⁴⁷

3.3 The Text-to-Text Framework: A Unified Denoising Objective

The Text-to-Text Transfer Transformer (T5) introduced a unified framework that reframes every NLP task as a text-to-text problem.⁵⁷ Instead of having different model architectures or output formats for different tasks (e.g., a classification head for

sentiment analysis, a span prediction head for question answering), T5 treats every task as a process of generating a target text string from a given input text string.³⁰

The pretraining objective for T5 is a generalized **denoising objective** that resembles a "fill-in-the-blank" task. The mechanism works as follows:

1. **Input Corruption:** Contiguous spans of tokens are randomly selected and dropped from the input text. Each dropped span is replaced by a single unique sentinel token (e.g., <X>, <Y>, etc.).⁵⁸
2. **Reconstruction Objective:** The model, which uses a standard Transformer encoder-decoder architecture, is trained to predict the original content of the dropped spans. The target output sequence consists of the sentinel tokens followed by the text they replaced.⁵⁹ For example, if the input is "Thank you for inviting me to your party last week," it might be corrupted to "Thank you <X> me to your party <Y> week." The model would then be trained to generate the target string: "<X> for inviting <Y> last <Z>" (where <Z> marks the end).⁵⁸

To adapt the pretrained T5 model to a specific downstream task, a short textual prefix is added to the input to instruct the model on what to do. For example, to perform translation, the input would be "translate English to German: That is good," and the model would be trained to generate "Das ist gut".⁵⁷ This elegant approach demonstrated that a single model and objective could achieve state-of-the-art results across a diverse suite of NLP benchmarks, including translation, question answering, and text classification, showcasing the power and flexibility of the text-to-text paradigm.⁶⁰

3.4 Replaced Token Detection (RTD): Efficient Discriminative Pretraining

While MLM is highly effective, it is also computationally inefficient, as the model only learns from the small subset of tokens (around 15%) that are masked in each training instance.⁶² Replaced Token Detection (RTD), introduced by the ELECTRA model, was designed to address this inefficiency by reframing the pretraining task from a generative one to a more sample-efficient discriminative one.⁶²

The RTD mechanism employs a two-model setup: a small **generator** and a larger **discriminator**.⁶³

1. **Generator:** The generator is typically a small MLM-style model. It takes an input

sequence with some tokens masked and generates plausible predictions for those masked positions.⁶⁴

2. **Input Corruption:** The original input sequence is then "corrupted" by replacing the masked tokens not with a `` token, but with the tokens sampled from the generator's output distribution.⁶²
3. **Discriminator Objective:** The discriminator, which is the main ELECTRA model being pretrained, is then presented with this corrupted sequence. Its task is to perform a binary classification for *every single token* in the sequence: it must predict whether each token is "original" or has been "replaced" by the generator.⁶²

The key advantage of RTD is that the discriminator learns from all tokens in the input sequence, not just the masked 15%. This dense learning signal makes the pretraining process significantly more computationally efficient.⁶² As a result, ELECTRA models can achieve performance comparable to or even better than much larger MLM-based models like BERT, but with substantially less pretraining compute.⁶³

The choice of pretraining objective in NLP is deeply intertwined with the model's architecture and its intended downstream applications. There is a clear coupling between the task and the architecture that optimizes for it. BERT's encoder-only architecture is a natural fit for the MLM objective, as the encoder's ability to see the entire input sequence is necessary to create deep bidirectional representations. This structure excels at NLU tasks that require a holistic understanding of a sentence, such as classification or token labeling.⁵⁴ Conversely, GPT's decoder-only architecture is intrinsically linked to the autoregressive CLM objective. This structure is designed for sequential generation, making it the ideal choice for tasks like open-ended text creation, dialogue, and summarization.⁴⁵ T5's encoder-decoder architecture provides a middle ground, capable of both deep input understanding (encoder) and flexible sequence generation (decoder), which perfectly complements its unified text-to-text framework for sequence-to-sequence tasks like translation.⁵⁷ This demonstrates that there is no universally superior architecture; rather, the choice reflects a strategic trade-off between understanding, generation, and versatile mapping, with the pretraining task tailored to exploit the strengths of the selected design.

Furthermore, the evolution of these tasks reveals a persistent drive for greater computational and sample efficiency. While MLM was revolutionary, its core inefficiency—learning from only 15% of the input tokens per sample—presented a clear bottleneck.⁶² The development of ELECTRA and its RTD objective was a direct response to this challenge. By reframing the task from a generative one ("predict the correct word") to a discriminative one ("is this word plausible?"), the model is able to

derive a learning signal from every token in the sequence.⁶⁴ This 100% token utilization makes the pretraining process far more efficient, allowing ELECTRA to match the performance of larger models with a fraction of the computational cost.⁶³ This progression underscores a critical trend in pretraining research: gains are achieved not just by scaling up models and data, but also by designing more intelligent and efficient learning objectives. As the cost of pretraining continues to rise, this quest for efficiency remains a central driver of innovation in the field.⁶⁶

4. Pretraining Tasks in Computer Vision

The pretraining paradigm in computer vision has undergone a dramatic transformation over the past decade. It has transitioned from a long-standing reliance on supervised classification on labeled datasets to a vibrant and diverse ecosystem of self-supervised methods that learn from unlabeled images at a massive scale. This evolution has not only improved performance on downstream tasks but has also addressed fundamental limitations related to data annotation and model generalization.

4.1 The Legacy and Limitations of Supervised Pretraining

For many years, the standard protocol for developing high-performance computer vision models was to pretrain them on the ImageNet-1k dataset using a supervised image classification objective.¹⁶ This approach proved to be exceptionally effective, as the hierarchical features learned by the model—from simple edges and textures in early layers to complex object parts in later layers—were highly transferable to a wide variety of other vision tasks.¹⁴ This ImageNet pretraining became a ubiquitous starting point that improved model performance and accelerated convergence for applications ranging from object detection to semantic segmentation.

However, this long-dominant paradigm has been increasingly challenged on several fronts. First, research has demonstrated that its benefits are not universal. On large-scale downstream datasets like COCO, models trained from scratch with strong data augmentation can match or even exceed the performance of their

ImageNet-pretrained counterparts, suggesting that the value of pretraining diminishes when sufficient downstream data is available.¹⁶ Second, the ImageNet dataset itself has faced scrutiny for significant ethical issues, including the use of images without consent, inherent societal biases reflected in its labels, and unclear copyright status.³¹ These concerns have created a strong impetus to develop pretraining methods that do not rely on large, manually curated, and potentially problematic labeled datasets, leading to the rise of self-supervised learning.³¹

4.2 Contrastive Learning Frameworks: Learning by Comparing

Contrastive learning has emerged as one of the most powerful families of self-supervised methods for visual representation learning. The core principle is to train a model to distinguish between similar and dissimilar pairs of images in a learned embedding space.¹⁸ In this framework, "positive pairs" are typically created by applying two different random data augmentations (e.g., cropping, color jittering, rotation) to the same source image. "Negative pairs" consist of augmented views from different source images.³⁷ The model is then trained with a contrastive loss to maximize the similarity of positive pairs while minimizing the similarity of negative pairs.¹⁸

4.2.1 SimCLR (A Simple Framework for Contrastive Learning)

SimCLR presented a simplified yet highly effective framework for contrastive learning.³⁸ Its mechanism involves taking an image, creating two strongly augmented views of it (the positive pair), and then passing both through an encoder network (e.g., a ResNet) followed by a small MLP projection head. The objective is to maximize the cosine similarity between the projected representations of the positive pair, while treating all other augmented images within the same large training batch as negative examples.⁴⁸ The seminal SimCLR paper made several key findings that became foundational for subsequent work: (1) the composition of strong data augmentations is critical for defining an effective learning task; (2) adding a learnable nonlinear projection head between the encoder's representation and the contrastive loss significantly improves feature quality; and (3) the method benefits immensely from very large batch sizes, which provide a large number of diverse negative examples for

comparison.³⁸

4.2.2 MoCo (Momentum Contrast)

The requirement of extremely large batch sizes for SimCLR posed a practical challenge, as it demanded significant GPU memory. Momentum Contrast (MoCo) was developed to address this limitation.³⁹ MoCo introduces a dynamic dictionary of negative samples, implemented as a queue. This queue stores the encoded representations from the most recent mini-batches, allowing the dictionary size to be decoupled from the batch size and thus be much larger.³⁹

A key challenge with this approach is ensuring that the keys in the dictionary are encoded consistently over time, as the encoder network is constantly being updated. To solve this, MoCo employs two encoders: a query encoder that is updated via standard backpropagation, and a key encoder that is not. Instead, the key encoder's weights are updated as a slow-moving momentum-based average of the query encoder's weights.⁷⁰ This momentum update ensures that the key encoder evolves smoothly, maintaining the consistency of the representations in the dictionary queue. MoCo demonstrated that it was possible to achieve state-of-the-art results without the need for massive batch sizes, making high-performance contrastive learning more accessible.³⁹ Later versions, like MoCo v2, incorporated successful elements from SimCLR, such as the MLP projection head and stronger augmentations, further improving performance.⁷⁰

4.3 Masked Image Modeling (MIM): The BERT of Vision

Directly inspired by the success of Masked Language Modeling in NLP, Masked Image Modeling (MIM) has emerged as a powerful generative alternative to contrastive learning.⁴⁹ This approach is particularly well-suited for Vision Transformers (ViTs), which naturally process images as a sequence of patches, analogous to tokens in a sentence. The general idea is to mask out a portion of the input image patches and train the model to predict the missing content.

4.3.1 MAE (Masked Autoencoders): Reconstructing Pixels

Masked Autoencoders (MAE) presented a simple, efficient, and highly scalable implementation of the MIM concept.⁴² Its design is based on two core principles:

1. **A High Masking Ratio:** MAE masks a very large proportion of the input image patches—typically 75%—using a random sampling strategy. This forces the model to learn a holistic, non-local understanding of the image, as it cannot simply rely on adjacent patches to infer the missing content.⁴²
2. **An Asymmetric Encoder-Decoder Architecture:** This is a key architectural innovation. The main, high-capacity encoder network operates *only* on the small subset of visible, unmasked patches (e.g., 25% of the total). A separate, much smaller and more lightweight decoder then takes the encoded representations of the visible patches, along with learned mask tokens representing the missing positions, and reconstructs the full image in the pixel space.⁴²

This asymmetric design makes the pretraining process extremely computationally efficient. By processing only a fraction of the patches in the powerful encoder, MAE can be trained much faster and on larger models than methods that process the full image. The learned representations have been shown to transfer exceptionally well, particularly for dense prediction tasks like semantic segmentation that benefit from a rich, holistic visual understanding.⁴²

4.3.2 BEiT (Bidirectional Encoder representation from Image Transformers): Predicting Visual Tokens

BEiT proposed a variant of MIM where the prediction target is not raw pixels but discrete "visual tokens".⁷⁶ The pretraining process involves two stages:

1. **Image Tokenization:** An offline, pre-trained visual tokenizer (specifically, the VQ-VAE from OpenAI's DALL-E) is used to convert each image into a sequence of discrete visual tokens from a predefined codebook.
2. **Masked Prediction:** Similar to MAE, a portion of the input image patches are masked. The BEiT model, a standard ViT, is then trained to predict the original discrete visual token corresponding to each masked patch.⁷⁶

By framing the reconstruction task as a classification problem over a discrete

vocabulary, BEiT draws a direct analogy to BERT's masked language modeling. BEiT was one of the first self-supervised methods to demonstrate that a pretrained Vision Transformer could outperform a model pretrained with supervision on the ImageNet-1k classification task, solidifying the power of the MIM paradigm.⁷⁶

4.4 Generative and Auxiliary Pretext Tasks

Before the dominance of contrastive learning and MIM, the field of vision SSL was pioneered by a variety of clever pretext tasks that derived supervision from the intrinsic properties of images.¹⁸ While often surpassed by modern methods in terms of raw performance, these tasks established the core principles of self-supervision. Key examples include:

- **Image Inpainting (Context Encoders):** The model is trained to fill in a missing region (often a central square) of an image. To do this plausibly, the model must learn about the structure of objects and the statistics of natural textures.⁴³
- **Image Colorization:** The model receives a grayscale version of an image and is trained to predict its original color channels. This requires a semantic understanding of the scene, as the model must learn that objects like "sky" are typically blue and "grass" is typically green, without being explicitly told what those objects are.⁴³
- **Predicting Geometric Transformations:** This category includes tasks that manipulate the geometry of the image. In **rotation prediction**, the model must predict which of four rotation angles (0°, 90°, 180°, 270°) was applied to an input image. In **jigsaw puzzles**, an image is divided into patches, which are then shuffled, and the model must predict their correct relative arrangement. These tasks force the model to learn about object orientation, parts, and spatial relationships.¹⁸

The following table provides a comparative analysis of the leading self-supervised vision pretraining methods, highlighting their key differences in mechanism and design.

Method	Learning Paradigm	Core Objective	Handling of Negatives	Key Architectural Innovation	Strengths

SimCLR	Contrastive	Maximize agreement of augmented views	Requires very large batch for negative samples	MLP projection head	Excellent linear probe performance , simple concept
MoCo	Contrastive	Maximize agreement of augmented views	Uses a momentum queue/dictionary for negatives	Momentum encoder & queue	Memory-efficient contrastive learning, no large batch needed
MAE	Generative (Reconstructive)	Reconstruct raw pixels of masked patches	Not applicable (non-contrastive)	Asymmetric encoder-decoder	Highly efficient pretraining, strong fine-tuning for dense tasks
BEiT	Generative (Predictive)	Predict discrete visual tokens of masked patches	Not applicable (non-contrastive)	Use of an external visual tokenizer	Strong fine-tuning performance , first to beat supervised pretraining

The evolution of self-supervised methods in vision can be understood as a search for different and increasingly effective ways to solve the fundamental problem of "representational collapse," where a model learns a trivial solution (e.g., outputting the same constant vector for every input). Contrastive methods like SimCLR and MoCo address this explicitly through **repulsion**; the loss function directly pushes the representations of negative samples apart, preventing them from all collapsing to a single point.³⁹ The primary engineering challenge then becomes how to source a large and diverse set of negative examples efficiently, leading to the innovations of large batches (SimCLR) and the momentum queue (MoCo). In contrast, generative MIM methods like MAE solve the collapse problem implicitly through the difficulty of the

reconstruction objective. A collapsed, low-information representation would be insufficient to reconstruct a complex, high-variance signal like a partially masked image. The necessity of predicting the missing pixels forces the encoder to preserve

rich, high-dimensional information, thereby preventing collapse without any explicit negative sampling.⁴² This reveals that the diversity of successful SSL methods can be viewed as different solutions to the same core challenge, with the choice reflecting a preference for either explicit contrastive pressure or the implicit pressure of a challenging generative task.

This transition from supervised to self-supervised learning also represents a fundamental shift in the resource economy of AI development. Supervised pretraining on ImageNet was a **curation-intensive** process, requiring a massive, one-time investment of human labor to collect, clean, and meticulously label over a million images.³¹ Self-supervised learning, by design, eliminates this human annotation bottleneck, enabling the use of vast, unlabeled, and often noisy web-scale datasets.¹⁸ However, to extract a meaningful learning signal from this raw data, SSL methods are inherently more

computation-intensive, requiring larger models, more training epochs, and often specialized techniques like large batch training or momentum encoders.³⁸ This trade-off—exchanging expensive human annotation for scalable compute—has been overwhelmingly successful. It has enabled the use of datasets orders of magnitude larger than ImageNet, which has been a primary driver of the performance and generalization capabilities of modern vision foundation models.³¹

5. Pretraining Tasks in Speech and Audio Processing

Self-supervised pretraining has fundamentally reshaped the field of speech and audio processing, enabling the development of powerful models that can learn the intricate structure of speech from vast quantities of unlabeled audio data. The primary challenge in this domain is the nature of the data itself: speech is a continuous, high-resolution waveform without the explicit, discrete units (like words or characters) found in text.⁴⁶ The most successful pretraining tasks have therefore focused on innovative ways to adapt the "mask-and-predict" paradigm from NLP to this continuous signal, typically by first learning a discrete representation of the speech units.

5.1 The Challenge of Speech Representation Learning

The goal of speech pretraining is to produce representations that are useful for a wide range of downstream tasks, including Automatic Speech Recognition (ASR), speaker identification, emotion recognition, and source separation.⁸² To do this, the model must learn to disentangle and encode various layers of information present in the raw audio, such as phonetic content (what is being said), speaker identity (who is speaking), and prosody (how it is being said). The continuous and unsegmented nature of the audio signal makes this a non-trivial problem for self-supervised methods.⁴⁶ Early approaches that attempted to predict low-level spectral features directly were often challenged by the continuous and high-variance nature of these targets.⁸²

5.2 wav2vec 2.0: Contrastive Learning on Quantized Latent Representations

The wav2vec 2.0 framework was a landmark achievement in self-supervised speech learning, demonstrating that a model pretrained on unlabeled audio could achieve state-of-the-art ASR performance when fine-tuned on just a few minutes of transcribed data.⁸⁴ It achieved this by combining ideas from convolutional feature extraction, Transformers, and contrastive learning over a learned, quantized representation space.⁴¹

The architecture and pretraining mechanism of wav2vec 2.0 consists of several key components:

1. **Feature Encoder:** A multi-layer 1D Convolutional Neural Network (CNN) takes the raw audio waveform as input and produces a sequence of latent audio representations at a lower temporal resolution (e.g., one vector every 25ms).⁸¹ This encoder learns to capture local acoustic features.
2. **Quantization Module:** The model jointly learns a codebook of discrete speech units. The latent representations from the CNN are mapped to these discrete units. This step is crucial as it discretizes the continuous audio signal, creating a finite set of targets for the prediction task.⁸⁷
3. **Masking:** In the latent space (before quantization), spans of the feature encoder's output are randomly masked, similar to the strategy used in BERT.⁴¹
4. **Contextualizer (Transformer):** A standard Transformer encoder takes the

sequence of masked latent representations as input and builds powerful, contextualized representations that integrate information from across the entire utterance.

5. **Contrastive Objective:** For each masked time step, the model's contextualized output is used to solve a contrastive task. The objective is to identify the correct quantized representation of the masked latent vector from a set of distractors, which are other quantized representations sampled from the same utterance.⁴¹ This forces the model to learn representations that are predictive of the underlying discrete speech units.

By pretraining on thousands of hours of unlabeled speech, wav2vec 2.0 learns representations that are highly effective for ASR, dramatically reducing the reliance on large, transcribed datasets and revolutionizing the approach to building speech systems for low-resource languages.⁸¹

5.3 HuBERT: Masked Prediction of Discrete Hidden Units

The Hidden-Unit BERT (HuBERT) model builds upon the success of the mask-and-predict paradigm but simplifies the pretraining objective by decoupling the generation of target labels from the main training loop.⁴⁶ Instead of a jointly learned quantization module and a contrastive loss, HuBERT uses an offline clustering step to provide discrete target labels for a standard BERT-like prediction loss.⁸⁹

The HuBERT pretraining process is iterative:

1. **Label Generation (Iteration 1):** An offline k-means clustering algorithm is first applied to basic acoustic features (e.g., Mel-frequency cepstral coefficients, or MFCCs) extracted from the entire unlabeled training dataset. This process generates a set of C cluster centroids, and each 25ms frame of audio is assigned a discrete cluster ID from 1 to C . These cluster IDs serve as the initial set of "hidden unit" pseudo-labels.⁹⁰
2. **Masked Prediction:** The raw audio is fed into a feature encoder and then masked, just as in wav2vec 2.0. A Transformer model is then trained with a simple cross-entropy loss to predict the correct cluster ID for each masked time step.⁸⁹ Crucially, the loss is only applied over the masked regions, forcing the model to learn a combined acoustic and language model to infer the missing targets from the surrounding context.⁴⁶

3. **Iterative Refinement:** This is a key innovation of HuBERT. After the first-iteration model is trained, its learned representations (extracted from an intermediate layer of the Transformer) are of much higher quality than the initial MFCCs. These superior representations are then used as input for a second round of k-means clustering. This generates a new, more refined set of target cluster labels. A new HuBERT model is then trained from scratch using these improved labels.⁴⁶ This iterative process of improving representations, which in turn improves the target labels for the next iteration of training, can be repeated multiple times.

A core finding of the HuBERT work is that the success of the method relies more on the *consistency* of the unsupervised clustering step than on the absolute acoustic quality of the assigned labels.⁴⁶ As long as the clustering algorithm consistently assigns the same label to similar sounds, the Transformer can effectively learn the sequential structure of speech. HuBERT demonstrated performance that matched or surpassed wav2vec 2.0 on major ASR benchmarks, highlighting the power of a simple masked prediction objective when combined with iteratively refined discrete targets.⁴⁶ The framework has also been successfully extended to the audio-visual domain with models like AV-HuBERT.⁹³

The success of both wav2vec 2.0 and HuBERT reveals a critical insight for self-supervised learning on continuous data: the importance of **discretization**. Unlike text, which is inherently composed of discrete units (words or subwords), speech is a continuous waveform. Attempting to directly predict continuous spectral features in a generative task is a challenging regression problem that can be sensitive to minor variations and may not encourage the model to learn the abstract, categorical nature of phonetic units.⁸² Both wav2vec 2.0 and HuBERT solve this by first transforming the problem into a discrete domain. They introduce a finite vocabulary of "speech units" or "tokens" and reframe the pretraining objective as a task of predicting the correct discrete token for a given segment of audio. wav2vec 2.0 achieves this discretization online through a learned quantization module and a contrastive objective that discriminates between these quantized units.⁸⁷ HuBERT accomplishes it offline, using the k-means clustering algorithm to discover the discrete units and then framing the task as a simpler classification problem (masked prediction) over these discovered units.⁹⁰ This shared principle—that discretizing the target space is key to successfully applying the powerful mask-and-predict paradigm to a continuous signal—is a fundamental contribution of modern speech pretraining research.

Furthermore, the iterative refinement process in HuBERT exemplifies a powerful meta-learning strategy: creating a self-reinforcing loop between the quality of a model's representations and the quality of its training targets. The initial targets

generated from simple acoustic features like MFCCs are inherently noisy and imperfect.⁸⁹ However, training a large Transformer model to predict even these noisy targets forces it to learn meaningful representations that capture the underlying acoustic and linguistic structure of speech; the model effectively learns to "see through the noise." These newly learned representations are far superior to the initial MFCCs. When they are used as the basis for the next round of clustering, they produce a cleaner, more consistent, and more acoustically meaningful set of target labels.⁴⁶ Training a new model on these better targets leads to even better representations, creating a virtuous cycle of improvement. This "bootstrapping" approach demonstrates that a model's own learned representations can be used to generate superior training data for itself, providing a highly effective method for enhancing performance without any external supervision and highlighting the robustness of the masked prediction objective to noisy targets.

6. Pretraining Tasks in Multimodal Learning

Multimodal pretraining aims to build models that can understand and reason about information from multiple data modalities simultaneously. The most prominent and successful subfield of this research area is vision-language learning, which focuses on connecting images and text. The central goal of pretraining in this domain is **alignment**: learning a shared, unified representation space where concepts from different modalities can be directly compared and related.⁹⁵ An effective alignment allows a model to understand that the pixels composing an image of a dog are semantically equivalent to the text string "a photo of a dog," enabling a host of powerful cross-modal applications.

6.1 The Goal of Multimodal Pretraining: Alignment

The primary challenge in multimodal learning is to bridge the "modality gap"—the inherent difference in the statistical properties and structures of different data types.⁹⁵ Pretraining tasks are designed to overcome this gap by learning a joint embedding space. In this space, an image vector and the text vector describing that image should be close to one another, while vectors from unrelated image-text pairs

should be far apart. Achieving this alignment is the key to unlocking critical downstream capabilities such as:

- **Cross-Modal Retrieval:** Searching for images using a text query (text-to-image retrieval) or finding relevant captions for a given image (image-to-text retrieval).⁹⁷
- **Visual Question Answering (VQA):** Answering a natural language question about the content of an image.⁹⁹
- **Image Captioning:** Generating a descriptive text caption for a given image.

6.2 Vision-Language Contrastive (VLC) Learning

Vision-Language Contrastive (VLC) learning has become the dominant and most scalable paradigm for pretraining large-scale vision-language models.¹⁰⁰ This approach applies the principles of contrastive learning to massive datasets of image-text pairs, which are often sourced directly from the web.¹⁰²

The VLC mechanism is characterized by a few key components:

1. **Dual-Encoder Architecture:** The model typically consists of two independent encoders: an image encoder (e.g., a Vision Transformer or a ResNet) to process visual data and a text encoder (e.g., a Transformer) to process language data.¹⁰³ These encoders map their respective inputs into the shared embedding space.
2. **Large-Scale Web Data:** VLC models are pretrained on enormous datasets containing hundreds of millions or even billions of (image, text) pairs scraped from the internet. The text often comes from the "alt-text" HTML attribute associated with images.¹⁰⁴
3. **Contrastive Objective:** The learning objective is to correctly associate images with their corresponding captions within a large training batch. For a batch of N image-text pairs, the model computes an $N \times N$ similarity matrix containing the cosine similarity between every image embedding and every text embedding. The model is then trained to maximize the similarity scores for the N correct pairs (positive pairs) located on the diagonal of this matrix, while simultaneously minimizing the similarity scores for the $N^2 - N$ incorrect pairings (negative pairs) on the off-diagonal.¹⁰¹

This simple yet powerful framework was popularized by two seminal models that demonstrated its effectiveness at an unprecedented scale.

6.2.1 CLIP (Contrastive Language-Image Pre-training)

Developed by OpenAI, CLIP demonstrated the remarkable capabilities that emerge from applying the VLC framework to a large, diverse dataset of 400 million image-text pairs from the web.⁴⁰ The most significant outcome of CLIP was its powerful

zero-shot classification ability.⁴⁰ After pretraining, CLIP can classify images into arbitrary categories without having been explicitly trained on any examples from those categories. This is achieved by transforming the classification task into an image-text matching problem. For a given image and a set of possible classes, text prompts are created for each class (e.g., "a photo of a cat," "a photo of an airplane"). The model then encodes the image and all the text prompts and predicts the class whose prompt has the highest cosine similarity with the image embedding.⁴⁰ This flexibility allows CLIP to generalize to a vast range of visual concepts far beyond the scope of traditional fixed-category classifiers like those trained on ImageNet.

6.2.2 ALIGN (A Large-scale Image and Noisy-text embedding)

Developed by Google, ALIGN pushed the scaling of VLC even further by pretraining on a massive and noisy dataset of 1.8 billion image-alt-text pairs.¹⁰⁸ A key insight from the ALIGN work was that the sheer scale of the dataset could compensate for a high degree of noise in the labels.¹⁰⁴ Unlike previous datasets like Conceptual Captions, which underwent extensive filtering and cleaning, the ALIGN dataset was created with only minimal filtering.¹⁰² By demonstrating that a simple dual-encoder architecture with a contrastive loss could learn state-of-the-art representations from this noisy, web-scale data, ALIGN proved that massive data scale could be a viable alternative to meticulous data curation, thereby significantly lowering the barrier to creating powerful vision-language models.¹⁰⁴

6.3 Other Multimodal Pretraining Tasks

While VLC is the dominant approach, other pretraining tasks are often used in conjunction with it or in different architectural setups to encourage a tighter fusion between modalities. These include:

- **Image-Text Matching (ITM):** This is a binary classification task where a model is given an image and a text caption and must predict whether the pair is a positive match (the text describes the image) or a negative match. This objective explicitly trains the model to assess the alignment of a given pair and is a common auxiliary task in many vision-language models.⁹⁷
- **Masked Language Modeling with Visual Context (MLM-VC):** This task is an extension of the standard MLM objective. When predicting a masked word in a caption, the model is allowed to attend not only to the surrounding text but also to the features from the corresponding image. This forces the model to learn fine-grained relationships between specific words and visual regions, promoting a deeper level of cross-modal fusion.⁹⁷

The success of models like CLIP and ALIGN marks a significant strategic victory for architectural simplicity and massive scale over model complexity. Before their emergence, many leading vision-language models employed complex "fused" architectures. In these models, image region features and text tokens were combined early in the network and processed jointly through multiple layers of cross-modal attention, with the goal of learning fine-grained, token-level interactions.⁹⁷ While effective on smaller, curated datasets like MS-COCO, these fused architectures are computationally expensive and difficult to scale. CLIP and ALIGN, in contrast, adopted a much simpler dual-encoder architecture where the two modalities are processed independently until their final representations are compared with a simple dot product.¹⁰³ This architectural simplicity is far more computationally efficient, which is precisely what enabled training to be scaled up to billions of noisy image-text pairs. The key discovery was that the general, robust representations that emerged from this massive-scale training with a simple objective were more powerful and transferable than those learned from smaller-scale training with more complex, fused architectures.¹⁰⁴ This suggests that for learning general-purpose multimodal foundations, the scale and diversity of weakly supervised data can be more impactful than intricate, task-specific architectural designs.

Furthermore, the nature of the web-based training data fundamentally shapes the knowledge these models acquire. The image-text pairs, derived from sources like alt-text, do not constitute a formal, structured ontology. Instead, they represent a vast, user-generated "folksonomy"—a reflection of how ordinary people describe visual scenes in natural, everyday language.¹⁰⁴ The contrastive pretraining objective

forces the model to align visual patterns with this rich and diverse linguistic tapestry. This is why CLIP can perform zero-shot classification on an open vocabulary of concepts, not just a fixed set of 1000 nouns like a traditional ImageNet model.⁴⁰ However, this global alignment comes with a trade-off. The model learns at the level of the entire image and the entire caption; it understands that a particular image corresponds to the text "a person riding a horse on a beach," but it does not explicitly learn which pixels correspond to the "horse" and which correspond to the "beach." This results in a well-documented limitation: while CLIP possesses strong image-level semantic understanding, its ability to perform fine-grained localization or reason about the spatial relationships between multiple objects is comparatively weak.¹⁰⁰ This demonstrates how the pretraining task directly molds the model's capabilities. Global image-text contrastive learning yields robust but holistic representations, which has in turn motivated subsequent research into new pretraining objectives, such as region-level contrastive losses, designed to explicitly instill a more fine-grained, localized understanding.¹⁰⁰

7. Pretraining in Specialized and Emerging Domains

While the principles of pretraining have been most prominently demonstrated in the domains of language, vision, and audio, they are increasingly being adapted to more specialized and structured data types. This section explores the application of pretraining to graph-structured data and the dynamic, interactive world of reinforcement learning and robotics. In these domains, pretraining is used not only to learn general representations but also to tackle unique challenges such as non-Euclidean data structures and the extreme sample inefficiency of real-world interaction.

7.1 Graph Representation Learning

Graphs are a universal data structure used to model relationships in a wide variety of domains, from social networks and knowledge bases to molecular structures and transportation systems. Learning on this non-Euclidean data requires specialized architectures, most notably Graph Neural Networks (GNNs).¹¹¹ Pretraining for GNNs

aims to learn transferable knowledge about graph topology and node/edge features from large, unlabeled graph datasets, which can then be applied to downstream tasks like node classification or link prediction.¹¹²

A compelling example of specialized graph pretraining is **GraphMVP (Graph Multi-View Pre-training)**, a framework designed for molecular graph representation learning.¹¹³ This domain presents a unique opportunity for multi-view learning, as molecules can be represented in two complementary ways:

1. **A 2D Topological View:** A standard graph where atoms are nodes and chemical bonds are edges. This view is readily available for most molecules.
2. **A 3D Geometric View:** The molecule's 3D spatial conformation, which provides rich information about its shape and energy state. This view is more predictive of molecular function but is often computationally expensive to obtain or unavailable.¹¹³

GraphMVP leverages these two views in a pretraining phase to distill the rich information from the 3D view into a standard 2D GNN encoder. This is achieved through a combination of two self-supervised tasks¹¹³:

- **Contrastive SSL:** The framework learns to align the representations of the 2D and 3D views of the *same* molecule (treating them as a positive pair) while contrasting them with the views of *different* molecules (negative pairs).
- **Generative SSL:** The framework also learns to reconstruct the representation of one view from the representation of the other, forcing the encoders to capture the essential information required to translate between the topological and geometric domains.

After this multi-view pretraining, the resulting 2D GNN encoder, now imbued with implicit knowledge of 3D geometry, can be fine-tuned on downstream tasks where only the 2D graph data is available. This approach has been shown to significantly outperform methods pretrained on 2D data alone, demonstrating the power of using pretraining to fuse information from data sources of varying richness and availability.¹¹³

7.2 Reinforcement Learning (RL) and Robotics

Pretraining in reinforcement learning and robotics is motivated by the desire to

overcome the severe sample inefficiency and safety concerns associated with training agents through trial-and-error, especially in the real world.¹¹⁵ The goal is to leverage existing data—from simulations, human demonstrations, or prior robotic experiences—to provide the agent with a foundation of knowledge before it begins online interaction and learning in a new environment.¹¹⁷

Several distinct pretraining strategies have emerged in this domain:

- **Offline RL Pretraining:** This approach involves pretraining a policy on a large, static dataset of previously collected experiences (state-action-reward-next-state tuples) without any further interaction with the environment. This is a powerful paradigm for leveraging existing robotic datasets. A key technical challenge is mitigating the "distribution shift" that occurs when the pretrained policy, during online fine-tuning, takes actions that lead it to states not well-represented in the offline dataset.¹¹⁶
- **Representation Learning from Action-Free Video:** This strategy focuses on pretraining a model to understand the dynamics of the physical world from large, unlabeled video datasets (e.g., videos of humans performing tasks) that contain no explicit action information.¹¹⁵ The framework **APV (Action-Free Pre-training from Videos)** exemplifies this approach. It pretrains a generative model to predict future video frames in a latent space. This task-agnostic "world model" learns a general understanding of physics and object interactions. To adapt this knowledge to a specific robot, a small, action-conditional prediction model is stacked on top of the frozen pretrained model during fine-tuning. This efficiently adapts the general knowledge of world dynamics to the specific embodiment and action space of the robot.¹¹⁵
- **Pretraining Rewards and Policies with Vision-Language Models:** A major bottleneck in real-world RL is the need for manual reward engineering. Recent work has shown that large, pretrained Vision-Language Models (VLMs) like CLIP can be used to provide autonomous reward signals. For a new task defined by a natural language goal (e.g., "pick up the red block"), the VLM can be used to create a reward classifier that evaluates whether the robot's current state (an image) matches the goal description. This allows a pretrained policy to be fine-tuned on a new task with minimal human intervention, as the reward function is generated automatically.¹¹⁶

The application of pretraining in these specialized domains reveals a powerful underlying theme: the use of pretraining to fuse "**privileged information**" into a model that operates on a more common or readily available data modality. In molecular science, the 3D geometric structure of a molecule is a form of privileged

information—it is highly informative but not always available at test time. The GraphMVP framework is explicitly designed to distill this privileged 3D knowledge into a standard 2D graph encoder during pretraining.¹¹³ Similarly, in robotics, a full trajectory that includes the robot's internal states and motor commands is a privileged view of a task. A simple video of the task being performed is a more widely available but less informative, "action-free" view. Pretraining from action-free video is a strategy to learn a general model of world dynamics that can then be rapidly specialized when the privileged action information becomes available during fine-tuning.¹¹⁵ This demonstrates that multi-view and cross-modal pretraining is a general and powerful principle for knowledge transfer, applicable not only to aligning human-centric modalities like image and text but also to bridging data sources of varying richness and accessibility in scientific and engineering contexts.

Furthermore, the trajectory of pretraining in reinforcement learning points towards a fundamental shift in objective. Early approaches focused on direct policy transfer: pretraining a policy on one set of tasks and then fine-tuning it on a new, similar task.¹¹⁷ However, more recent and ambitious methods, particularly those leveraging video pretraining, are not learning a policy directly. Instead, they are learning a predictive model of how the world works—a

world model.¹¹⁵ This world model captures the physics and dynamics of the environment in a task-agnostic manner. Once learned, this internal simulator can be used by a planning algorithm (such as model-predictive control) to derive a policy for a

new task at test time, often with little to no additional training. The agent can use its world model to "imagine" the consequences of different action sequences and choose the one that best achieves its goal. This represents a move away from learning *what to do* (a policy) and towards learning *how the world works* (a model). This more general and powerful form of knowledge is central to building more flexible, adaptable, and intelligent agents, and it provides a direct bridge to the future directions of pretraining discussed in the final chapter.

8. The Evolution and Future of Pretraining

The pretraining paradigm has evolved from a niche technique for improving model initialization into the central, defining process for creating the most powerful and

general AI systems today. This survey has charted its course across diverse modalities, revealing a set of consistent evolutionary trends that point toward an ambitious future. This concluding chapter synthesizes these key trends and looks ahead to the next frontier of pretraining: the development of comprehensive world models that can understand, simulate, and ultimately act within complex environments.

8.1 Synthesis of Key Evolutionary Trends

Across the domains of language, vision, audio, and beyond, the evolution of pretraining has been characterized by several overarching themes:

- **From Supervised to Self-Supervised:** The most fundamental shift has been the move away from reliance on costly, human-labeled datasets towards self-supervised learning on vast, unlabeled web-scale data. This transition was the catalyst that enabled the creation of modern foundation models.²²
- **The Dominance of Scaling Laws:** A key empirical discovery has been the predictable, power-law relationship between performance and the scale of model size, dataset size, and training compute. This has fueled a relentless drive toward building ever-larger models trained on ever-larger datasets, as scale has consistently proven to be a reliable path to improved capabilities.⁴⁷
- **Architectural and Objective Convergence:** The Transformer has emerged as a near-universal architecture, capable of processing diverse data types as sequences of tokens. This has been accompanied by a cross-pollination of pretraining objectives, with core ideas like masked prediction and contrastive learning being successfully adapted across modalities, suggesting the existence of fundamental principles of representation learning.⁴⁷
- **Democratization through the Pretrain-Finetune Split:** The immense cost of pretraining has led to a bifurcation of the AI development ecosystem. A small number of organizations perform the industrial-scale pretraining, while a much broader community of researchers and developers can access and fine-tune these powerful foundation models for specific applications. This has democratized access to state-of-the-art AI capabilities.²⁵

8.2 The Next Frontier: Pretraining World Models

While most pretraining tasks discussed in this survey focus on learning passive representations of data, the next frontier aims to create models that can actively reason, plan, and make decisions. This ambition is coalescing around the concept of the **world model**.¹¹⁹

A world model is an internal, generative model that learns to represent and simulate an environment.¹¹⁹ It captures the dynamics of the world—the rules, physics, and causal relationships that govern how states evolve over time and in response to actions. By learning such a model, an AI agent can effectively "dream" or "imagine" the future, allowing it to predict the consequences of its actions and plan a course of action to achieve a goal without costly or dangerous real-world trial-and-error.¹¹⁹ This capability is widely seen as a critical stepping stone toward more general and autonomous AI systems.¹²²

The pretraining objective for a world model is fundamentally predictive and often generative. The core task is, given a history of observations and actions, to predict future observations.¹²¹ To make this tractable, the prediction is often performed in a compressed latent space rather than directly in the high-dimensional pixel space.¹¹⁹ Recent research, such as Meta's

V-JEPA 2, exemplifies this approach. V-JEPA 2 is a world model pretrained on over a million hours of video data to learn a predictive model of the physical world. This pretraining is "actionless," meaning it learns the natural dynamics of the world from passive observation. The resulting model can then be efficiently adapted with a small amount of robot-specific, action-conditioned data to enable planning and control for embodied agents.¹²¹ This two-stage process—large-scale pretraining for general world understanding followed by lightweight adaptation for specific embodiment—bridges the gap between passive representation learning and active, goal-directed behavior.¹⁰

8.3 Concluding Remarks

Pretraining has matured from a simple initialization trick into the engine of modern AI. The field is propelled by a dual pursuit: on one hand, the relentless scaling of models and data to capitalize on empirical scaling laws, and on the other, the clever design of

more efficient and intelligent self-supervised objectives that make this scaling feasible. The trajectory of this evolution points toward a clear and ambitious end-game. The initial pretrain-finetune paradigm, which assumed that task-specific fine-tuning was a necessary final step, is gradually giving way to a new ideal.²³ The success of in-context learning in LLMs like GPT-3 and the zero-shot capabilities of vision-language models like CLIP demonstrate a powerful trend: as pretraining becomes more effective and is conducted at a larger scale, the need for extensive, gradient-based fine-tuning diminishes.⁴⁰ The ultimate goal is to create pretrained models that are so general and capable that they can be instructed to perform new tasks on the fly, with the "tuning" process becoming a lightweight act of prompting or goal specification rather than a heavyweight retraining process. The world model represents the apotheosis of this trend for agentic AI; a sufficiently powerful world model would not need to be retrained for every new task but could be used by a planner to derive novel solutions dynamically.¹²¹

This progression also suggests a deeper shift in the nature of the knowledge being learned. Most current pretraining tasks are designed to learn statistical correlations from observational data. They learn that "lightning" is often followed by "thunder," but they lack a deep, causal model of the underlying atmospheric physics. World models, particularly those conditioned on actions, represent a crucial step toward learning **causal relationships**.¹²¹ They must learn the causal link between an agent's intervention and the subsequent change in the environment's state. This is a far more robust and powerful form of knowledge than mere correlation, as it enables true reasoning about interventions ("what will happen if I do this?") and counterfactuals ("what would have happened if I had done that differently?"), which are the hallmarks of genuine intelligence and are essential for robust planning.¹¹⁹ The future of pretraining will therefore likely involve a greater focus on learning from interventional and dynamic data sources, such as robotics interaction logs and complex simulations. The objective will evolve from simply predicting missing data to predicting the consequences of actions—a fundamentally causal pretraining task. This will be the critical path toward building AI systems that can safely, effectively, and intelligently interact with the real world.

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